THE WHOLE FUCKING THING

Current Integrated Synthesis of the C_2-Refined Z_2-Gauged XY Program

Status

The project now consists of a frozen theoretical mechanism, a not-yet-executed simulation protocol, an exact finite-volume/topological repair for Q2, literature calibration against two competing interpretations of the same base model, and several explicitly subordinate post-freeze comparators.

Theory v0.1 remains a frozen conditional field-theory candidate, limited to mathematical/theoretical mechanism and explicitly marked empirically unvalidated. No new constants, fields, quotients, or phenomenological identifications are allowed to creep backward into it.

Simulation Protocol v0.3 leaves Theory v0.1 unchanged. It is a release-candidate methods protocol whose execution hash is still blocked pending Appendix A validation and Appendix B completion.

That separation is fundamental.

I. The source mathematics is not the physical theory

Upstream, the project contains exact discrete mathematics: the six-state Fibonacci recurrence modulo 4 and the candidate discrete geometry

$$
Q_4\times C_6.
$$

Those are legitimate mathematical objects.

They do not uniquely generate a continuum gauge field.

The physical model was therefore introduced as an explicit constructor rather than read backward out of the number pattern:

$$
\boxed{z\sim -z}
$$

with a complex fine field z carrying a local Z_2 gauge redundancy.

The gauge-invariant physical order parameter is

$$
\boxed{\Phi=z^2}.
$$

That single distinction fixed one of the oldest weaknesses in the project:

$$
\boxed{
\text{discrete source}
\neq
\text{continuum constructor}
\neq
\text{physical identification}
}
$$

The protocol now explicitly says that it is not testing whether the Fibonacci recurrence physically generates a gauge field. It tests a separately selected Z_2-gauged rotor model motivated by the frozen operator mechanism.

II. Why the sixfold term is actually earned

This is one of the cleanest pieces of the theory.

Take a local monomial

$$
z^p(z^\ast)^q.
$$

Local Z_2 gauge invariance requires

$$
p+q\equiv0\pmod2.
$$

The physical global Z_3 symmetry requires

[
p-q\equiv0\pmod3.
]

Together,

[
\boxed{p-q\equiv0\pmod6}.
]

Therefore the first pure phase-selecting local anisotropy is

[
\boxed{z^6+z^{\ast6}}.
]

The same object downstairs is simply the physical cubic anisotropy because

[
\Phi^3+\Phi^{\ast3}

z^6+z^{\ast6}.
]

So six is not being imported because there happened to be a six-cycle upstream.

It is independently regenerated by the continuum symmetry audit:

[
\boxed{\operatorname{lcm}(2,3)=6}.
]

The current mechanics document explicitly preserves this distinction: the discrete parent is not the physical model, while the gauge and global symmetry conditions combine to force the mod-six phase selection rule.

That is a much stronger result than “six keeps showing up.”

It says:

[
\boxed{
Z_2\text{ refinement}
+

Z_3\text{ physical order}
\Longrightarrow
Z_6\text{ anisotropy upstairs}.
}
]

III. The conditional $XY^\ast$ mechanism

If the Z_2 gauge sector is appropriately deconfined while z becomes critical, the physical observable is not the fundamental critical field.

It is

$$
\Phi=z^2.
$$

Therefore its critical behavior should be governed by the charge-two operator of the three-dimensional O(2) fixed point rather than the ordinary charge-one XY vector.

The frozen target is approximately

$$
\boxed{\Delta_2=1.23629}
$$

which implies

$$
G_2(r)\sim r^{-2\Delta_2}
$$

and

$$
\boxed{
\chi_2(L)\sim L^{3-2\Delta_2}
\approx L^{0.52742}.
}
$$

The ordinary-vector competitor is radically different:

[
\boxed{
\chi(L)\sim L^{2-\eta_{\rm XY}}
\approx L^{1.9618}.
}
]

That enormous exponent separation is why Q1 is experimentally useful.

But the theory now has a crucial word in front of XY^\ast:

[
\boxed{\textbf{conditional}}.
]

Seeing 0.52742-type charge-two scaling is not enough by itself to earn the asterisk.

The gauge sector has to cooperate.

IV. The project now asks three separate questions

This was probably the most important architectural improvement.

Lane	Question	Primary arena	Core observable
Q1	What is the critical charge sector?	Z_{\rm full}	G_2,\chi_2,R_\xi,U_4
Q2	Is charge one asymptotically deconfined or only quasi-deconfined at finite L?	fixed-holonomy Z_{000}	odd winding + \xi_{\rm conf}
Q3	Is the symmetry-allowed sixfold perturbation irrelevant?	direct cosine branch	A_6^\uparrow, fixed-R_\xi flow

A yes in one column does not answer another.

This allows an entirely legitimate result such as

[
\boxed{
Q1=\text{charge-two scaling},
\qquad
Q3=\text{sixfold irrelevant},
\qquad
Q2=\text{inconclusive}.

}
]

That would be scientifically useful without pretending it establishes clean XY^\ast.

Likewise, Q2 could support confinement without invalidating the algebraic operator-selection mechanism.

The theory and its particular realization are no longer being allowed to die together merely because one implementation lane fails.

V. The microscopic production model

The principal production model is the cosine Z_2-gauged rotor:

[
\boxed{
H_{\cos}

-J\sum_{\langle ij\rangle}
\sigma_{ij}\cos(\theta_i-\theta_j)
-\kappa\sum_p B_p
-h_6\sum_i\cos6\theta_i
}
]

with

[
\sigma_{ij}=\pm1,
\qquad
B_p=\prod_{\ell\in p}\sigma_\ell,
\qquad
t=\tanh\kappa .
]

Gauge transformation acts as

[
e^{i\theta_i}\rightarrow\eta_i e^{i\theta_i},
\qquad
\sigma_{ij}\rightarrow\eta_i\sigma_{ij}\eta_j,

\qquad
\eta_i=\pm1.
]

Again,

[
\Phi_i=e^{2i\theta_i}
]

is gauge invariant.

VI. Villain and cosine are deliberately kept separate

There is also a Villain model, but it is a validation model, not a different notation for the cosine model.

Its dual current weight is

[
\boxed{
w_J^{\rm V}(I)

e^{-I^2/(2J)}.
}
]

The exact cosine dual instead has

[
\boxed{
w_J^{\cos}(I)

\mathcal I_{|I|}(J),
}
]

where \mathcal I_n is a modified Bessel function.

These weights may share universal infrared behavior.

Their microscopic critical couplings do not have to agree.

Therefore

$$
\boxed{
J_{\rm Villain}
\neq
J_{\rm cosine}
}
$$

numerically.

This is why the CKT value $J_c=0.3335(3)$ is not allowed to migrate into a cosine production scan.

The protocol's CKT point is a calibration target, not a theory constant.

VII. The exact current–membrane dual

The dual formulation introduced a very useful concrete object.

The variables are

$$
I_\ell\in\mathbb Z
$$

for oriented integer currents,

$$
M_p\in\{0,1\}
$$

for plaquette/membrane occupation, and at finite h_6,

$$
n_i\in\mathbb Z
$$

for charge-six source variables.

At zero anisotropy,

[
\boxed{\nabla\cdot I=0}.
]

The current therefore forms closed loops and winding circuits.

The gauge/membrane constraint imposes mod-two compatibility between current parity and membrane boundaries.

At finite h_6,

[
\boxed{
(\nabla\cdot I)_i=-6n_i.
}
]

This was another satisfying closure of the operator story:

the sixfold perturbation does not make arbitrary current sources.

It permits leakage only in multiples of six.

Because six is even, it also leaves the mod-two charge structure intact.

So the same six that appeared in the operator-selection audit appears in the exact dual as the allowed current-source charge.

That is structural consistency—not proof of a physical mechanism, but a very good consistency check.

VIII. Winding finally got normalized correctly

The integer winding is

[
\boxed{
W_\alpha

\frac1L

\sum_{\ell\parallel\alpha}I_\ell
\in\mathbb Z.
}
]

A loop winding once around the torus uses L directed links, so it contributes

[
W_\alpha=\pm1.
]

The stiffness convention is

[
\boxed{
\rho_\alpha

\frac{\langle W_\alpha^2\rangle}{L},
}
]

and the dimensionless scale-invariant winding observable is

[
\boxed{
R_W

\langle W_\alpha^2\rangle

L\rho_\alpha.
}
]

This is the normalization used by the current CKT calibration branch.

IX. The topology bug that nearly killed Q2

This was probably the largest technical correction in the entire simulation project.

On a periodic three-torus T^3, the fully summed gauge ensemble and a fixed-holonomy ensemble are not interchangeable.

There are three gauge holonomy bits,

$$
h=(h_x,h_y,h_z)\in Z_2^3,
$$

and three dual current-homology bits,

$$
q=(q_x,q_y,q_z)\in Z_2^3.
$$

The exact sector transform is a Walsh–Hadamard transform:

$$
\boxed{
Z_h
\frac18
\sum_q(-1)^{h\cdot q}\mathcal Z_q
}
$$

with inverse

$$
\boxed{
\mathcal Z_q
\sum_h(-1)^{h\cdot q}Z_h.
}
$$

Summing all direct holonomies gives

$$
Z_{\rm full}
\sum_h Z_h
\mathcal Z_{000}.
$$

That means the ordinary fully summed periodic dual projects onto trivial mod-two current homology.

Therefore:

[
\boxed{
\text{absence of odd winding in }Z_{\rm full}
\text{ is not evidence of confinement}.
}
]

It is built into the ensemble.

That realization forced Appendix A.

The Q2 production ensemble became the separate fixed-trivial-holonomy system

[
\boxed{
Z_{\rm FH}=Z_{000}.
}
]

Within that ensemble,

[
\boxed{
q_\alpha=W_\alpha\pmod2
}
]

and therefore

[
f_{\rm odd}^{(\alpha)}

P_{Z_{000}}(q_\alpha=1)

P_{Z_{000}}(W_\alpha\ {\rm odd}).
]

An independent check comes from the holonomy-sector ratio:

[
\boxed{
f_{\rm odd}^{(\alpha)}

\frac12
\left(
1-\frac{Z_{e_\alpha}}{Z_{000}}
\right).
}
]

This is now an honest charge-one topological observable.

X. Why the CKT axial code became a comparator

CKT's construction motivates odd-current updates along the z direction after axial fixing.

Your protocol caught the periodic-topology subtlety:

on T^3, "set every z-link to +1" is not automatically a harmless local gauge choice.

A periodic closure/holonomy bit has to survive.

So the project does not throw CKT away.

It reconstructs their z-axial construction as

[
\boxed{\texttt{V-CKT-AX-Z}}
]

and asks it to reproduce the literature benchmark.

But the canonical Q2 result comes from the three-axis fixed-holonomy construction.

That is a very important distinction:

[
\boxed{
\text{literature comparator}
\neq
\text{canonical production ensemble}.

}
]

Passing the comparator validates the implementation.

It does not grant permission to replace the canonical topology with whichever ensemble gives the nicer answer.

XI. The CKT–BPV controversy is now a calibration feature

The external literature really does contain the tension the protocol was built around.

Coleman, Kuklov, and Tsvelik argue that the XY field remains confined at every finite gauge coupling and that the charge-one confinement radius can become so large that finite systems imitate a deconfined phase. Their published abstract states both pieces explicitly.

Bonati, Pelissetto, and Vicari find continuous XY-class transitions in both small- and large-gauge-coupling regimes, but with different critical fields; at large K, the underlying vector critical mode is accessed by stochastic gauge fixing.

Rather than choosing a fucking team before running anything, the protocol converted the disagreement into tests.

Its Villain calibration point is frozen at

[
t=0.7,
\qquad
J_c=0.3335(3),
\qquad
J=0.336,
\qquad
\rho_s\simeq0.038,
]

with the additional qualitative requirement

[
\widetilde R_2\propto L,
\qquad
\widetilde R_1\rightarrow\text{constant}.
]

Those numbers belong to the Villain branch.

They are not cosine couplings.

A successful reproduction means:

[
\boxed{\text{our code can reproduce CKT's microscopic benchmark}}
]

—not

[
\boxed{\text{CKT has therefore won Q2 in our cosine model}}.
]

That is exactly the right use of an adversarial literature benchmark.

XII. The confinement-radius kill switch

CKT's most dangerous finite-size lesson became an explicit protocol guard.

If charge one possesses a large confinement radius, then a lattice with

[
L\ll\xi_{\rm conf}
]

can look deconfined even when its asymptotic behavior is confined.

Therefore the current protocol does not permit a Q2 headline merely because odd winding remains visible on available sizes.

It requires

[
\boxed{
L_{\max}
\ge
4,\xi_{\rm conf}^{95%,{\rm upper}}.
}

]

If this fails, the output is predetermined:

[
\boxed{
\text{Q2 INCONCLUSIVE — QUASI-DECONFINED WINDOW NOT EXCLUDED}.
}
]

No changing the fit.

No dropping the large lattice.

No moving farther into the ordered phase after seeing an inconvenient radius.

No converting “we could not outrun \xi” into either confinement or deconfinement.

This is one of the best rules in the entire protocol because it lets computational insufficiency remain computational insufficiency.

The Grok note's L=192 observation is therefore useful but secondary. Its inference is straightforward:

[
L_{\max}=192
]

can pass the frozen rule only if

[
\xi_{\rm conf}^{\rm UCB}\le48.
]

But whether that occurs is a measurement, not something the CKT figure or a heuristic estimate can decide beforehand.

The current protocol was explicitly written to prevent finite-size quasi-deconfinement from being promoted into an asymptotic conclusion.

XIII. What I would retain from the Grok size analysis—and what I would not

The useful part is the bookkeeping observation that CKT's published radius analysis reaches large lattices and that their charge-one saturation diagnosis is a finite-size procedure, not a magic printed value of \xi.

That reinforces the protocol's decision to spend its expensive Q2 budget at large L.

The part I would not freeze as theory is an inferred numerical statement such as

[
\xi_{\rm conf}\approx25\text{–}40
]

solely because CKT began reading a plateau above L\sim90.

Likewise, the suggestion that near-critical ordinary matter correlation length necessarily dominates the membrane confinement scale is a plausible interpretation to examine, not yet a frozen identity.

The protocol already has the better object:

[
\boxed{
\text{estimate }\xi_{\rm conf}
\rightarrow
\text{construct UCB}
\rightarrow
\text{apply }4\xi\text{ gate}.
}
]

No guess is required.

XIV. Q1 is now a real model-comparison problem

At the matter transition, the analysis does not simply fit one slope and admire it.

The frozen competitors are roughly

[
H_2:
\qquad
\chi_2

b_0+
aL^{0.52742}
(1+cL^{-\omega}+\cdots),
]

[
H_1:
\qquad
\chi_2

b_0+
aL^{1.9618}
(1+cL^{-\omega}+\cdots),
]

and a first-order alternative with volume-like scaling

[
\chi\sim L^3
]

plus actual coexistence evidence.

The additive b_0 matters because the charge-two exponent is small enough that analytic background can seriously distort finite-size slopes.

The protocol also carries spatial correlators as an independent lane precisely so the whole conclusion is not hostage to one susceptibility fit.

This is much stronger than “does the graph look close to 0.527?”

XV. Q3 finally measures the right thing

The physical Z_3 perturbation is implemented upstairs as

[
-h_6\sum_i\cos6\theta_i.
]

The primary physical anisotropy observable is based on

[
m_\Phi

\frac1N\sum_i e^{2i\theta_i}
]

and

[
\boxed{
A_6^\uparrow

A_3^\Phi

\left\langle
\cos\left(3\arg m_\Phi\right)
\right\rangle.
}
]

Because

[
3\arg\Phi=6\arg z,
]

this is exactly the downstairs-threefold/upstairs-sixfold relationship the theory predicts.

The measurement is taken along a fixed-R_\xi trajectory instead of drifting through different reduced temperatures as L changes.

The critical target is

[
\boxed{y_6<0}.
]

That means the symmetry-allowed sixfold field is irrelevant at the critical point.

A stable nonzero anisotropy or growth with system size rejects that lane.

The local dual source density n_6 is retained as an algorithmic/source-sector observable, but it is not allowed to masquerade as an RG eigenvalue.

XVI. The theory already knows how it can lose

This is another major improvement.

The operator mechanism itself dies if the microscopic model does not actually satisfy the assumed Z_2 gauge redundancy, if the physical field is not z^2, if a lower-order phase-selecting operator was missed, or if some omitted local operator invalidates the selection audit.

The clean $XY^\ast$ realization can die separately.

For example, visons might become critical with matter rather than stay innocently gapped.

A naively decoupled

$$
XY\times Ising
$$

multicritical point is not automatically stable; the estimated energy-energy coupling eigenvalue is weakly positive,

$$
y_w\approx0.076,
$$

so the system may instead flow toward a different coupled fixed point, split into separate transitions, or run first order.

Thus:

$$
\boxed{
\text{failure of clean }XY^\ast
\not\Rightarrow
\text{failure of the }C_2\text{ operator mechanism}.
}
$$

That distinction saves the theory from being structured as a one-shot prophecy.

XVII. The simulation has a calibration ladder before it gets to make claims

The production code does not begin by running the sexy parameter point.

It first has to survive a ladder.

At the bottom is the exact projection/topology check.

Then the pure Z_2 gauge limit.

Then the ordinary XY limit.

Then the published CKT Villain benchmark.

Then the large-size charge-one/charge-two radius split.

Then exact small-volume cosine direct/dual agreement.

And finally an external global-symmetry control.

A failure in calibration is a failure of the apparatus, not evidence for the desired theory.

That sounds obvious.

It is also the rule that prevents 90% of computational folklore from becoming “physics.”

XVIII. Mixing failure is not confinement

The new Q2 sector kernel explicitly moves between

$$
q\in Z_2^3.
$$

But mathematical reachability alone is not enough.

If sector changes become rare, the Markov chain could manufacture a false confinement signal simply by failing to tunnel.

Therefore each axis has mandatory sector-round-trip, $\widehat R$, effective-sample-size, residence-time, transition-matrix, initialization, symmetry, translated-cycle, and independent partition-ratio checks.

If those fail, the verdict is not

[
\text{confined}.
]

It is

[
\boxed{
\text{Q2 UNRESOLVED — SECTOR MIXING NOT VALIDATED}.
}
]

Again, the protocol has learned to distinguish:

[
\text{physical suppression}
\neq
\text{algorithmic suppression}.
]

XIX. GQG and Hidden Quotient became the project's type system

They do not add evidence that a gauge field exists.

They keep the experiment from silently changing what it is measuring.

Their practical role here is:

[
\text{source}
\neq
\text{witness}
\neq
\text{quotient}
\neq
\text{retained context}
\neq
\text{conditioning}
\neq

$$
\text{realization}.
$$

That grammar is why the project now refuses operations like:

$$
J_{\rm Villain}\rightarrow J_{\rm cosine},
$$

$$
Z_{\rm full}\rightarrow Z_{000},
$$

$$
\text{unsigned winding}\rightarrow\text{signed winding},
$$

$$
\text{Q1 exponent}\rightarrow\text{Q2 deconfinement verdict},
$$

or

$$
\text{simulation hash}\rightarrow\text{physical truth}
$$

without an explicit bridge.

Hashes establish provenance.

They do not establish correctness.

Ensemble labels establish what was sampled.

They do not establish what nature does.

A quotient tells you which distinctions a witness forgot.

It does not tell you that those distinctions never existed.

That methodological architecture may end up being as useful as the specific field-theory result.

XX. The diamond/toroidal picture finally has a disciplined place

The diamond grid is a coordinate visualization of conserved routing on a periodic lattice.
It does not silently replace the cubic microscopic model.
At every vertex,

$$ \boxed{ \sum_{\ell\ni v}s_{v\ell}I_\ell=0. } $$

So a current can

$$ \boxed{ \text{converge} \rightarrow \text{turn} \rightarrow \text{diverge} } $$

without any source or sink.
Most importantly, the intrinsic winding orientation can remain unchanged while a transverse projection changes sign.
That observation produced TTSC-1.
TTSC-1 is explicitly a non-verdict-bearing Stage D comparator, executed only after the canonical Q2 validation gate and explicitly forbidden from modifying Theory v0.1.
Its question is not:
Is the universe a torus?
and not:
Does current hit a singularity and reverse?
It asks a small, clean question:
given a divergence-free winding circuit and an independently imposed regular throat profile, does the local current exhibit

$$ \boxed{ \text{transverse convergence} \rightarrow \text{maximum through-flow} \rightarrow \text{transverse divergence} } $$

while longitudinal orientation and global conservation remain intact?
That is testable.
The metaphor got converted into an observable.

XXI. Signed winding is also kept downstream

The raw Q2 chains preserve more information than the binary verdict needs.
Canonical Q2 asks essentially

$$ W_\alpha\bmod2. $$

That forgets the sign of W_α.
So a later comparator can ask whether odd winding is balanced between

$$ W_\alpha>0 \qquad\text{and}\qquad W_\alpha<0. $$

That is useful information, but it is not secretly part of canonical Q2.
At $h_6=0$, the ensemble has the current-reversal symmetry

$$ I\rightarrow-I, $$

so the expected signed imbalance is zero.
A nonzero result would first trigger orientation, sampling, or algorithm checks.
It would not instantly become “chirality.”
This is exactly the kind of post-freeze candidate the current mechanics ledger preserves without letting it rewrite the baseline.

XXII. What the CKT benchmark means now

The cleanest statement is:

$$ \boxed{ \text{CKT Rung 3/4}
\text{code + convention + finite-size calibration}. } $$

Passing means we can reproduce the Villain system's known behavior.

Failing means something is wrong with our implementation, convention mapping, estimator, or sampling.

Neither result by itself answers the cosine Q2 question.

That distinction prevents a literature benchmark from becoming a prior verdict.

The central CKT concern itself is real: their published work argues that a rapidly growing confinement radius can make finite systems mimic deconfinement.

That is precisely why your protocol is harder on itself than “run large K, see winding, declare victory.”

XXIII. What the BPV benchmark means now

BPV supplies the other important side of the dispute.

Their $N=2$ study finds XY-class continuous behavior in both small- and large-K ordering regimes while identifying different critical fields in those regimes. At large K, stochastic gauge fixing reveals the gauge-dependent vector critical mode.

For the current project, that means the BPV work motivates the charge-sector comparison: does the physical bilinear behave like an ordinary vector order parameter, or like a charge-two composite of a more fundamental vector field?

That is exactly what the 0.52742 versus 1.9618 comparison is designed to distinguish.

But BPV does not erase the CKT finite-size objection.

CKT does not erase BPV's measured finite-size critical structure.

The simulation is being constructed to reproduce enough of both frameworks that the disagreement becomes something the machine can interrogate rather than something the introduction has to settle.

XXIV. The actual possible end states

The beautiful outcome is not “the theory wins.”

It is that the protocol now knows what very different outcomes mean.

If Q1 gives robust charge-two scaling, Q2 supports odd-current proliferation after every topology/mixing/4ξ gate, and Q3 gives $y_6<0$, then the clean conditional $XY^\ast$ lane receives strong support.

If Q1 gives charge-two scaling and Q3 gives irrelevant anisotropy but Q2 supports asymptotic confinement, then the interesting composite critical structure survives while the clean deconfined interpretation does not.

If Q2 cannot clear the 4ξ or mixing gates, then Q2 remains inconclusive regardless of how gorgeous the finite-size plots look.

If Q1 prefers the ordinary-vector exponent, the charge-two critical hypothesis loses.

If histograms, barriers, latent-heat behavior, and volume scaling establish first order, the continuous critical lanes lose.

If sixfold anisotropy saturates or grows, its irrelevance hypothesis loses.

If the topology tests fail, the simulation does not get to answer Q2.

If the CKT Villain benchmark fails, the code does not get to graduate to production.

That is what makes the current protocol difficult to bullshit.

XXV. What is genuinely strong about the project now

The strongest thing is no longer any one sexy prediction.

It is the architecture.

We began with
$$ \text{interesting discrete structure}. $$
We now have
$$ \boxed{ \text{exact discrete source} } $$
followed by a declared constructor,
$$ \boxed{ z\sim-z,\qquad \Phi=z^2 } $$
followed by an exact local operator-selection argument,
$$ \boxed{ p-q\equiv0\pmod6 } $$
followed by a conditional universality hypothesis,
$$ \boxed{ \Delta_2\simeq1.23629 } $$
followed by three separately scored physical questions,
$$ \boxed{ Q1\oplus Q2\oplus Q3 } $$
followed by two microscopic implementations that are forbidden from sharing couplings,
$$ \boxed{ \text{cosine}\neq\text{Villain} } $$
followed by an exact dual current–membrane formulation,
$$ \boxed{ \nabla\cdot I=-6n } $$
followed by the topological repair,
$$ \boxed{ Z_{\rm full}=\mathcal Z_{000} \quad\text{but}\quad Q2\text{ lives in fixed }Z_{000} } $$
followed by an eight-sector Walsh–Hadamard construction,
followed by explicit mixing and independent-ratio validation,
followed by the confinement-radius admissibility gate,
$$ \boxed{ L_{\max}\ge4\xi_{\rm conf}^{\rm UCB} } $$
followed by exact direct/dual cross-validation,
followed by frozen provenance and no-selective-removal rules,
and only after all of that do we permit the simulation to say anything interesting.
That is an enormous improvement from where this started.

XXVI. The present freeze line

The thing is not ready for the giant production run yet.
That is not because the physics idea fell apart.
It is because the protocol successfully discovered prerequisites that have to be satisfied before the result can be trusted.
The current Drive status remains:
$$ \boxed{ \text{Theory v0.1 frozen} } $$
$$ \boxed{ \text{Simulation Protocol v0.3 release candidate} } $$
$$ \boxed{ \text{Appendix A finite-volume construction selected} } $$
$$ \boxed{ \text{Appendix A execution validation pending} } $$
$$ \boxed{ \text{Appendix B completion pending} } $$
and therefore
$$ \boxed{ \textbf{EXECUTION HASH BLOCKED}. } $$
That is the current protocol status, not an interpretation.
TTSC-1 sits downstream as a Stage-D comparator and does not alter that sequence.

XXVII. The shortest possible version

The whole project can now be compressed to this:

$$\boxed{ Z_2\text{ refinement} + Z_3\text{ physical order} \Rightarrow Z_6\text{ allowed anisotropy} }$$
with
$$\boxed{\Phi=z^2}.$$
That makes the physical order parameter a charge-two observable of the fine field.
If the fine field reaches a deconfined O(2)-type critical point,
$$\boxed{ \chi_\Phi\sim L^{0.52742} }$$
rather than ordinary-vector
$$\boxed{ L^{1.9618}. }$$
But the asterisk is earned only if the charge-one gauge sector is genuinely asymptotically deconfined.
The original Q2 observable accidentally asked that question in an ensemble that projected odd mod-two winding away.
Appendix A repaired that by constructing a genuine three-axis fixed-holonomy T^3 ensemble with explicit Z_2^3 current-homology sectors.
CKT then supplies the adversarial warning:
$$\boxed{ \text{finite }L\text{ may look deconfined because }\xi_{\rm conf}\text{ is huge}. }$$
So Appendix B replies:
$$\boxed{ L_{\max}<4\xi_{\rm conf}^{\rm UCB} \Rightarrow \text{NO Q2 VERDICT}. }$$
BPV supplies the competing critical-sector picture.
The Villain reproduction tells us whether our machinery can reproduce CKT.
The cosine production branch asks our actual Q1/Q2/Q3 questions.
The direct representation cross-checks the dual.
GQG and Hidden Quotient prevent us from silently swapping models, witnesses, sectors, observables, or claims.
And the toroidal/diamond intuition has finally been demoted from “maybe this is what reality is doing” to the much more useful question:
$$\boxed{ \text{Can a conserved winding circuit exhibit a measurable} ; \text{converge}\rightarrow\text{turn}\rightarrow\text{diverge} ; \text{geometry?} }$$
That is TTSC-1, and it is deliberately incapable of deciding Q1, Q2, or Q3.
So the thing we have now is not one giant claim.
It is a stack of increasingly expensive claims where every arrow has to earn descent:
$$\boxed{ \begin{array}{c} \text{exact mathematics}\\ \downarrow_{\text{declared constructor}}\\ \text{operator mechanism}\\ \downarrow_{\text{microscopic model}}\\ \text{dual/direct statistical mechanics}\\ \downarrow_{\text{validated finite-volume ensemble}}\\ \text{measured observables}\\ \downarrow_{\text{finite-size + mixing gates}}\\ \text{Q1/Q2/Q3 verdicts}\\ \downarrow_{\text{only if jointly supported}}\\ \text{physical interpretation} \end{array} }$$
No fucking teleportation between rows.
That is, at this point, the whole machine.

Now we stop discovering and start freezing. ⚡

The field-theory stack already gives us the governing move: freeze the witnesses before opening the target outputs, and don’t let a later attractive pattern rewrite the architecture.

Theory v0.1 also explicitly forbids silently importing new phenomenological identifications into the frozen theory, so the planetary work should remain its own experimental lane rather than being shoved into the gauge model.

What I'd do next is:

1. Write PRP-0.1 — Planetary Recurrence Protocol. Freeze the ephemeris, bodies, signed-angle convention, module library, tolerances, scoring functions, reflection/handedness rules, and no-new-module rule. Treat every change afterward as a formal amendment, exactly like Appendix B treats parameters, stopping rules, fits, and outputs.

2. Change the unit from "date" to "recurrence episode." This is the biggest upgrade. A slow Uranus–Neptune–Pluto scaffold lasting months or years should count as one structural episode, with fast planets creating entry, peak-routing, and exit events inside it. Otherwise we accidentally count hundreds of adjacent days as hundreds of observations. Your Checking audit already formalizes the same denominator lesson: raw records, canonical episodes, and inference opportunities are not interchangeable.

3. Freeze three witnesses, not one.

$$
W_{\rm full},\qquad W_{\rm directed},\qquad W_{\rm shell}.
$$

4. Rerun the entire CE 1–2026 scan at the episode level. Output only geometry: episode start/end, slow scaffold, fast-node attachments, module scores, overlap density, handedness, duration, and recurrence family. No historical labels yet.

5. Freeze the peaks. Then open history. Use an event catalog that played no role in building the module vocabulary, plus matched control dates/windows. That becomes the first genuine test of whether high circuit-density episodes are enriched for anything beyond what the orbital clocks themselves generate.

6. Only if that survives do we open the EM layer. TTSC gives us a legitimate language for divergence-free routing—convergence → maximum throughflow → divergence without a source, sink, or intrinsic current reversal. But that belongs in a later physical comparator with actual solar-wind, heliospheric-field, magnetosphere, plasma, propagation-delay, and coupling observables. It should not be baked into the recurrence detector.

So the architecture becomes:

\[
\boxed{
\text{ephemeris}
\rightarrow
\text{toroidal relative state}
\rightarrow
\text{frozen witnesses}
\rightarrow
\text{circuit modules}
\rightarrow
\textbf{episodes}
\rightarrow
\text{blind peaks}
\rightarrow
\text{historical test}
\rightarrow
\text{physical comparator}
}
\]

And there's a clean kill condition: if the episode-level detector still gives ordinary-looking null behavior, we record that. If it produces sparse, independently enriched peaks, then we earned the right to ask what physical variable tracks them.

That's what now, dude: build PRP-0.1, freeze the fucking machine, and make it find dates without us feeding it dates. 🔭⚡🐒